

Problems for Identity Theory: Leibniz' Principle and Multiple Realization

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Leibniz' Principle or the Principle of the Indiscernibility of Identicals.

The mathematical principle of the 'indiscernibility of identicals' says that if X and Y are identical, then they have all their properties in common. When we say 'identical', we mean 'exactly the same'. For example: Sally is identical to the tallest person in the room. In that case, Sally must be the tallest person in the room.

Now, because they are identical (meaning exactly the same), if Sally has the property of an IQ of 175, so does the tallest person in the room; and if the tallest person in the room has the property of owning a Harley Davidson, so does Sally. For two things to be identical, they cannot differ in any way.

Leibniz' principle also applies to "types" of things. For example, the type 'water' is identical to the type 'H₂O'. If water boils at 100 Celsius, so does H₂O; and if H₂O conducts electricity, so does water.

So if, as the Identity Theorist claims, pain is identical to a specific type of brain activity, then every property of pain must also be a property of that brain activity. This means that if we find a property of pain which cannot be a property of that brain activity then they are not identical, and Identity Theory is proven false.

Now, all pain has a property which is definitively part of its experience: it *hurts*. This is the property we will focus on for the next section.

Multiple Realization: humans, animals, and potentially artificial intelligences might all experience pain, but the physical processes causing this pain could be quite different. So, if those physical processes of pain are all different, then how can they all share the same property of *feeling hurt*?

Let's agree that, for the sake of argument, in humans' pain is a unique part of our brain responding. Now, we can easily imagine animals with brains quite different from our own. Let's agree, again for the sake of argument, that squid have brains quite different from our own and lack that part of the brain which we use to respond to pain (This isn't at all implausible. The squid brain is very different from our own. From an evolutionary perspective, humans and squid are only very distantly related.)

So far, we have assumed only that squid don't have that human part of the brain. It seems quite likely, though, that they experience pain that hurts (or at least we have no good reasons to doubt that they can be in pain). Consequently, the identity theory is in trouble: if squid can lack that part of the brain but feel pain that has the property of hurting, then it cannot be the case that pain is identical to a specific type of brain activity.

This poses a serious challenge to the Identity Theory. If pain can be realised in different ways across different physical systems (i.e., different brain structures or even non-biological systems), then pain cannot be identical to one specific type of brain activity. In other words, if pain (a mental state) can have properties that are not shared by a specific type of brain activity (a physical state), then according to Leibniz's Principle, they cannot be identical. Thus, Multiple Realization suggests that mental states are more than just physical processes and cannot be fully explained by them.

NOTE TO LEXIE: Functionalism, however, suggests that 'pain' and 'feeling hurt' are both merely processes which function to detect injury. For the Functionalist, it doesn't matter if 'feeling hurt' is different for everyone/every species: it retains the same function.

However, while Functionalism can explain how different physical states can experience functionally similar mental states, it is extremely limited in how it can explain the properties of the mind.